



COMMONWEALTH OF AUSTRALIA

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FIT3080: Artificial Intelligence

Week 5 Probabilistic Reasoning

Topic 1: Probability

Topic 2: Bayesian Networks

Guest lecturer: Steven Mascaro

Slides adapted from those of Ingrid Zukerman, Ann Nicholson, as well as Stuart Russell, Andrew Moore, Dan Klein & Kevin Korb

Introductions

- Me – Steven Mascaro
- You - FIT3080 students
 - Experience with logic and conditional probability?
 - Who remembers learning Bayes' theorem?
 - Who has downloaded Netica?

Knowledge Representation

- How do we represent facts about the world?
- How do we reason about those facts?
- Some widely accepted formal calculi
 - Propositional logic
 - First-order logic
 - Probability calculus





FIT3080: Artificial Intelligence

Part I – Probability

Russell & Norvig Chapter 13

Lecturer: Steven Mascaro

Probability Topic: Outline

- Background:
 - Random variables and probabilistic inference
 - Probabilistic models
 - Joint, marginal and conditional distributions
- Inference by enumeration
- Product Rule, Chain Rule, Bayes' Rule
- Independence and conditional independence

These are things you need to know to understand Bayesian networks and how they work.

Reasoning under Uncertainty

- Uncertainty – the quality or state of being not clearly known (very common)
 - Distinguishes deductive knowledge from inductive belief
- Sources of uncertainty
 - Ignorance
 - Complexity
 - Physical randomness
 - Vagueness

Probability Calculus (I)

- Classic approach to uncertainty (origin: Pascal and Fermat)
- Definitions:
 - **Experiment** – (or trial) produces one of several possible **outcomes**
 - **Sample space** – the set of all possible outcomes of the experiment
 - **Event** – a subset of the sample space
 - **Random variable** – a variable whose value is determined by the outcome of an experiment
 - > e.g. $C=\{h,t\}$ for coin toss, where: heads $\rightarrow C=h$, and tails $\rightarrow C=t$
 - > Each value is associated with an event, and all the events cover the full sample space and are mutually exclusive
 - **Probability function** – a function that assigns a probability to every possible outcome of an experiment or value of a random variable
 - > e.g. heads $\rightarrow 0.5$, tails $\rightarrow 0.5$ or $\Pr(C=h) \rightarrow 0.5$, $\Pr(C=t) \rightarrow 0.5$



Random Variables

- Represent some aspect of the world about which we may have uncertainty
 - R = Is it raining?
 - D = How long will it take to drive to work?
 - L = Where am I?
- Denote with capital letters (or capitalize e.g. Rain)
- Random variables have domains (i.e. set of values)
 - R in $\{\text{true}, \text{false}\}$ (sometimes write as $\{r, \neg r\}$)
 - D in $[0, \infty)$
 - L in possible locations, maybe $\{(0,0), (0,1), \dots\}$

Probability Distributions

- Random variables have distributions that represent probabilities of value assignments

Pr(Temperature)

Temperature	Pr
warm	0.5
cold	0.5

Pr(Weather)

Weather	Pr
sunny	0.6
rain	0.1
fog	0.3

- A probability is a single number

$$\text{Pr(Weather=rain)} = 0.1 \quad \text{or} \quad \text{Pr(rain)} = 0.1$$

Probability Calculus (II)

- Kolmogorov's axioms for probabilities over an event space

- where $e_1, \dots, e_n$ are all the possible distinct events in space E

1. $\forall e_i \Pr(e_i) \geq 0$

The probability of any e_i in E must be at least 0.

2. $\sum e_i \Pr(e_i) = 1$

*All probabilities have to sum to 1.
(1 & 2 imply $\forall e_i \Pr(e_i) \leq 1$)*

3. $\forall e_i, e_j \subseteq E$

if $e_i \cap e_j = \emptyset$ then

$\Pr(e_i \vee e_j) = \Pr(e_i) + \Pr(e_j)$

*If e_i and e_j represent separate events
(no overlapping outcomes), the
probability of one OR the other (or
both) occurring is the sum of the two.*

Apply to probabilities over **random variables**, because they are probabilities ($\Rightarrow 1$) that are exhaustive ($\Rightarrow 2$) and mutually exclusive ($\Rightarrow 3$)



Joint Distributions

- A joint distribution over a set of random variables $X_1, \dots, X_n$ specifies a probability for each value assignment (or outcome):

$$\Pr(X_1=x_1, \dots, X_n=x_n) \text{ or } \Pr(x_1, \dots, x_n)$$

- Size of joint distribution of n variables with domain sizes d ? (i.e. how many rows?)
- Must obey probability axioms:
 - Non-negative
 - Sum to 1 (because exhaustive)
 - Can be summed (because mutually exclusive)
- For all but small distributions, impractical to write out

$\Pr(T, W)$

T	W	Pr
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

sums
to 1

Probabilistic Inference

- Probabilistic inference:
 - Compute a desired probability from other known probabilities (e.g. using joint distribution)
- We generally compute conditional probabilities
 - They represent an agent's beliefs given some evidence
 - E.g., Unconditional: $\Pr(\text{on time}) = 0.70$
 - Conditional: $\Pr(\text{on time} \mid \text{no accidents}) = 0.90$
- Probabilities change with new evidence:
 - Observing new evidence causes beliefs to be updated
 - E.g., $\Pr(\text{on time} \mid \text{no accidents}, 5 \text{ a.m.}) = 0.95$
 - $\Pr(\text{on time} \mid \text{no accidents}, 5 \text{ a.m.}, \text{raining}) = 0.80$

Probabilistic Models (I)

- Probabilistic models describe how (a portion of) the world works
- **Models are always simplifications**
 - May not account for every variable
 - May not account for all interactions between variables
 - “*All models are wrong; but some are useful.*”
 - George E. P. Box
- **What do we do with probabilistic models?**
 - Reason about:
 - > Unknown variables of interest (i.e., unknown states of the world)
 - > Given evidence (i.e., known states of the world)
 - Examples:
 - > explanation (diagnostic reasoning)
 - > prediction (causal reasoning)
 - > value of information

Probabilistic Models (II)

- A probabilistic model is a joint distribution over a set of variables

$$\Pr(X_1, X_2, \dots, X_n)$$

- Given a joint distribution, we can reason about unobserved variables given evidence
- General form of a query:

$$\Pr(X_q | e_1, \dots, e_k)$$

Stuff you care about $\xrightarrow{\quad}$ $\xleftarrow{\quad}$ *Stuff you already know*

- This kind of posterior distribution is also called the belief function of whichever agent uses this model

Events in a Joint Distribution

- From a joint distribution, we can calculate the probability of any event

$$\Pr(E) = \sum_{\{x_1, \dots, x_n\} \in E} \Pr(x_1, \dots, x_n)$$

- Probability that it is hot AND sunny
 - Probability that it is hot
 - Probability that it is hot OR sunny
- Typically, the events we care about are partial assignments, like $\Pr(T=\text{hot})$

$\Pr(T, W)$

T	W	Pr
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3



Marginal Distributions*

- Marginal distributions are derived tables that eliminate variables
- Marginalization (summing out): Combine collapsed rows by adding

$\Pr(T, W)$

Temp	Weather	Pr
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

$$\Pr(T) = \sum_{w \in \{sun, rain\}} \Pr(T, w)$$

$\Pr(T)$

T	Pr
hot	0.5
cold	0.5

$$\Pr(W) = \sum_{t \in \{hot, cold\}} \Pr(t, W)$$

$\Pr(W)$

W	Pr
sun	0.6
rain	0.4

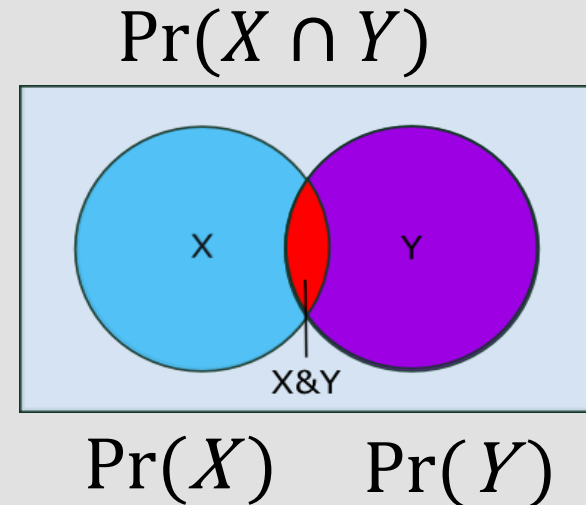


Conditional Distributions (I)

$$\Pr(X | Y) = \frac{\Pr(X \wedge Y)}{\Pr(Y)}$$

$\Pr(T, W)$

T	W	Pr
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

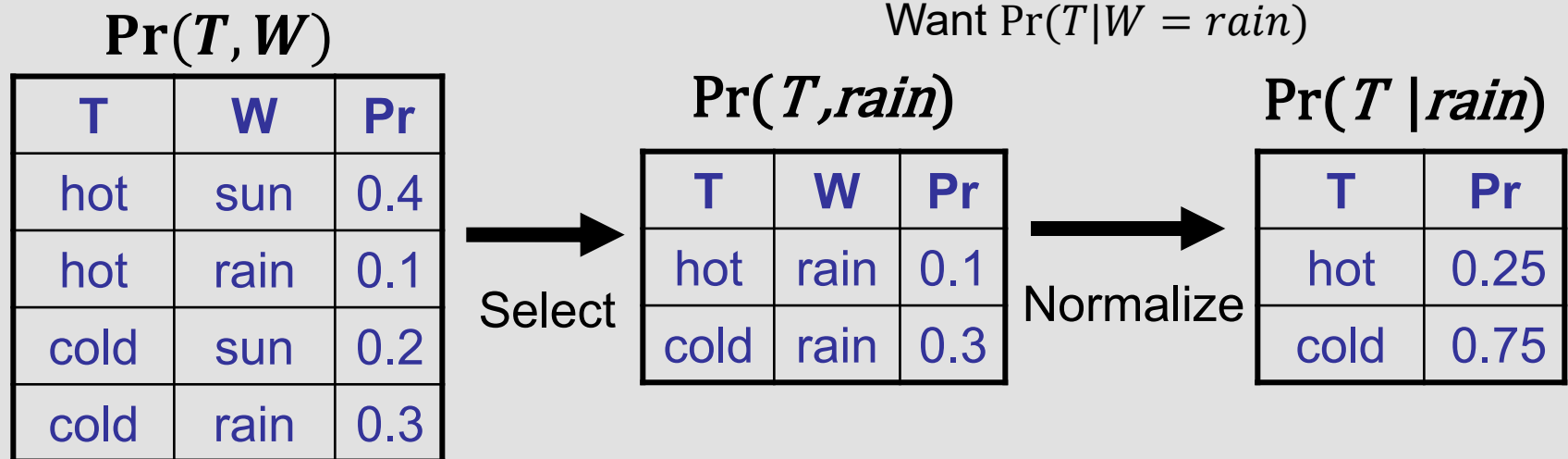


$$\Pr(W = \text{rain} | T = \text{cold}) = ?$$

$$\Pr(T = \text{cold} | W = \text{rain}) = ?$$

Normalization Trick

- A trick to get a conditional distribution:
 - Select the joint probabilities matching the evidence
 - **Normalize** the selection (make it sum to 1)



- Why does this work?

$$\Pr(T|r) = \frac{\Pr(T \wedge r)}{\Pr(r)} = \frac{\Pr(T \wedge r)}{\sum_{T=t} \Pr(t \wedge r)}$$

Conditional Distributions (II)

Can produce a *complete* conditional distribution $P(X|Y)$ from the joint distribution $P(X \wedge Y)$

Joint Distribution

$\Pr(T, W)$

T	W	Pr
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

Conditional Distributions

$\Pr(W|T)$

$\Pr(W|T = \textit{hot})$

W	Pr
sun	0.8
rain	0.2

$\Pr(W|T = \textit{cold})$

W	Pr
sun	0.4
rain	0.6



Inference by Enumeration (I)

May have many variables in our joint distribution, not just X & Y

- $\text{Pr}(\text{sun})?$
- $\text{Pr}(\text{sun} \mid \text{spring})?$
- $\text{Pr}(\text{sun} \mid \text{winter, hot})?$

Can compute these from the joint distribution, but tedious!

Season	Temp	Weather	Pr
spring	hot	sun	0.25
spring	hot	rain	0.05
spring	cold	sun	0.15
spring	cold	rain	0.05
winter	hot	sun	0.10
winter	hot	rain	0.05
winter	cold	sun	0.15
winter	cold	rain	0.20



Inference by Enumeration (II)

- General case:
 - Evidence variables: $E_1, \dots, E_k = e_1, \dots, e_k$
 - Query variable(s): Q
 - Unknown variables: $U_1, \dots, U_r$
- We want $\Pr(Q|e_1, \dots, e_k)$
- Procedure
 1. Select the entries that are consistent with the evidence
 2. Sum out U to get the joint probability of Query and Evidence:
$$\Pr(Q, e_1, \dots, e_k) = \sum_{u_1, \dots, u_r} \underbrace{\Pr(Q, u_1, \dots, u_r, e_1, \dots, e_k)}_{X_1, \dots, X_n}$$
 3. Normalize the remaining entries to conditionalize
- Problems:
 - Worst-case time and space complexity both $O(d^n)$
 - Issues with estimating the joint, particularly with small data

Inference by Enumeration – Example

- $\Pr(\text{sun} \mid \text{spring})$
 - Evidence variables?
 - Query variables?
 - Unknown variables?
- Procedure
 1. Select entries consistent with the evidence
 2. Sum out U to get a joint probability of Q and E

$$\Pr(\text{sun}, \text{spring}) = 0.25 + 0.15 = 0.4$$

$$\Pr(\text{rain}, \text{spring}) = 0.05 + 0.05 = 0.1$$

	S	T	W	Pr	
spring, sun	spring	hot	sun	0.25	spring, rain
	spring	hot	rain	0.05	
	spring	cold	sun	0.15	
	spring	cold	rain	0.05	
	winter	hot	sun	0.10	
	winter	hot	rain	0.05	
	winter	cold	sun	0.15	
	winter	cold	rain	0.20	

Inference by Enumeration – Example

- $\Pr(\text{sun} \mid \text{spring})$
 - Evidence variables?
 - Query variables?
 - Unknown variables?
- Procedure
 1. Select entries consistent with the evidence
 2. Sum out U to get a joint probability of Q and E
 3. Normalize the remaining entries to conditionalize

	S	T	W	Pr	
spring, sun	spring	hot	sun	0.25	spring, rain
	spring	hot	rain	0.05	
	spring	cold	sun	0.15	
	spring	cold	rain	0.05	
	winter	hot	sun	0.10	
	winter	hot	rain	0.05	
	winter	cold	sun	0.15	
	winter	cold	rain	0.20	

$$\begin{aligned}\Pr(\text{sun} \mid \text{spring}) &= 0.4 / (0.4 + 0.1) \\ &= 0.8\end{aligned}$$

$$\begin{aligned}\Pr(\text{sun}, \text{spring}) &= 0.25 + 0.15 = 0.4 \\ \Pr(\text{rain}, \text{spring}) &= 0.05 + 0.05 = 0.1\end{aligned}$$



The Product Rule

- Sometimes we have conditional distributions but want the joint distribution

$$\Pr(x|y) = \frac{\Pr(x, y)}{\Pr(y)} \quad \longleftrightarrow \quad \Pr(x, y) = \Pr(x|y) \Pr(y)$$

- Example:

$\Pr(W)$		$\Pr(T W)$			$\Pr(T, W)$		
W	Pr	T	W	Pr	T	W	Pr
sun	0.8	cold	sun	0.1	cold	sun	0.08
rain	0.2	hot	sun	0.9	hot	sun	0.72
		cold	rain	0.7	cold	rain	0.14
		hot	rain	0.3	hot	rain	0.06



The Chain Rule

- We can always write a joint distribution as an incremental product of conditional distributions

$$\Pr(x_1, \dots, x_n) = \prod_{i=1}^n \Pr(x_i | x_1, \dots, x_{i-1})$$

- Why is this true?
 - $\Pr(x_1, \dots, x_n) = \Pr(x_n | x_1, \dots, x_{n-1}) \Pr(x_1, \dots, x_{n-1})$
 $= \Pr(x_n | x_1, \dots, x_{n-1}) \Pr(x_{n-1} | x_1, \dots, x_{n-2}) \Pr(x_1, \dots, x_{n-2})$
- Example:
 $\Pr(\text{Traffic}, \text{Umbrella}, \text{Rain}) =$
 $\Pr(\text{Umbrella} | \text{Rain}, \text{Traffic}) \times \Pr(\text{Traffic} | \text{Rain}) \times \Pr(\text{Rain})$

Bayes Rule

- Two ways to factor a joint distribution over two variables:

$$\Pr(x, y) = \Pr(x|y) \Pr(y) = \Pr(y|x) \Pr(x)$$

$$\Pr(x|y) = \frac{\Pr(y|x) \Pr(x)}{\Pr(y)}$$

- Why is this helpful?
 - Lets us build one conditional from its reverse
 - Often one conditional is tricky but the other one is simple
 - > $\Pr(\text{Disease}|\text{Test Result})$ hard, $\Pr(\text{Test Result}|\text{Disease})$ easy
 - Foundation of many reasoning systems

Bayes Rule

- Attributed to Rev. Thomas Bayes

$$\Pr(h | e) = \frac{\Pr(e | h) \Pr(h)}{\Pr(e)}$$

- Also called *Conditionalization*:

$$\Pr'(h) = \Pr(h | e)$$

- Also read as

$$\text{Posterior} = \frac{\text{Likelihood} \times \text{Prior}}{\text{Prob of evidence}}$$

h = hypothesis
e = evidence

Rev. Thomas Bayes
(1702-1761)



Inference with Bayes Rule – Example

Probability of breast cancer (hypothesis), given xray test result (evidence)

Suppose that

- The xray returns a positive result
- 1 out of every 100 people who do the test have breast cancer

What do you think the probability of having breast cancer is?

I haven't given you enough information!

Inference with Bayes Rule – Example

Probability of breast cancer (hypothesis), given xray test result (evidence)

Suppose that

- The xray returns a positive result
- 1 out of every 100 people who do the test have breast cancer:
 $\Pr(h)=0.01$
- The true positive rate is 0.8: $\Pr(e|h)=0.8$
- The false positive rate is 0.1: $\Pr(e|\sim h)=0.1$

Now what do you think the probability is?

Inference with Bayes Rule – Example

Probability of breast cancer (hypothesis), given xray test result (evidence)

- Let $\Pr(h)=0.01$, $\Pr(e|h)=0.8$ and $\Pr(e|\sim h)=0.1$

$$\begin{aligned}\Pr(h|e) &= \frac{\Pr(e|h) \Pr(h)}{\Pr(e)} \\&= \frac{\Pr(e|h) \Pr(h)}{\Pr(e|h) \Pr(h) + \Pr(e|\sim h) \Pr(\sim h)} \\&= \frac{0.8 \times 0.01}{0.8 \times 0.01 + 0.1 \times 0.99} \\&= \frac{0.008}{0.008 + 0.099} = \frac{0.008}{0.107} \approx 0.075\end{aligned}$$

Posterior probability of breast cancer is still small. So should you still be alarmed if the xray is positive?



Conditional Independence

- X and Y are **independent** iff
$$\forall \mathbf{x}, \mathbf{y} \Pr(\mathbf{x}, \mathbf{y}) = \Pr(\mathbf{x})\Pr(\mathbf{y})$$
$$\Leftrightarrow X \perp\!\!\!\perp Y$$
- X and Y are **conditionally independent** given Z iff
$$\forall \mathbf{x}, \mathbf{y}, \mathbf{z} \Pr(\mathbf{x}, \mathbf{y}|\mathbf{z}) = \Pr(\mathbf{x}|\mathbf{z})\Pr(\mathbf{y}|\mathbf{z})$$
$$\Leftrightarrow X \perp\!\!\!\perp Y | Z$$
- (Conditional) independence is a property of a distribution

Conditional Independence

- Independence is a simplifying modeling assumption
 - Not common in empirical joint distributions, but “close” to independent quite common
- Conditional independence is our most basic and robust form of knowledge about uncertain environments
 - Examples:
 $\Pr(\text{Traffic}|\text{Umbrella}, \text{Rain}) = \Pr(\text{Traffic}|\text{Rain})$
(equivalent to $\Pr(\text{Traffic}, \text{Umbrella}|\text{Rain}) = \Pr(\text{Umbrella}|\text{Rain}) \times \Pr(\text{Traffic}|\text{Rain})$)
 - $\Pr(\text{alarm}|\text{fire}, \text{smoke}) = \Pr(\text{alarm}|\text{smoke})$
 - What could we assume for {Weather, Traffic, Cavity, Toothache}?
- Bayesian networks / graphical models help us express conditional independence assumptions

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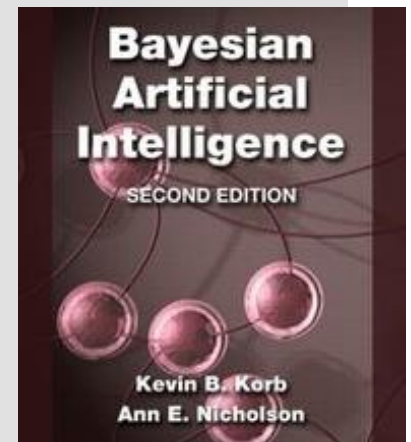
Bayesian Networks I: Representation and Reasoning

Russell & Norvig Chapters 14.1-3

Korb & Nicholson, 2011: Chapter 2

Lecturer: Steven Mascaro

Slides are adapted from those of Ann Nicholson, Stuart Russell, Andrew Moore, Dan Klein & Kevin Korb



BNs Part I: Learning Objectives

- Be able to:
 - Construct a syntactically correct BN that plausibly represents a problem given via an English language description
 - > i.e. Nodes, arcs & parameters (CPTs)
 - Calculate the joint probability distribution from a BN
 - > i.e. using the CPTs
 - Identify whether variables are conditionally dependent or independent for common BN fragments
 - Use the Netica BN software to build a BN and reason using that BN [lab and assignment]

Assumptions about the Environment

- Partially/fully observable
- Known/unknown
- Single/multi agent
- Deterministic/stochastic
- Episodic/sequential
- Static/dynamic
- Discrete/continuous

All can be relaxed (some with extensions),
but underlined is what we'll mostly cover

Bayesian Networks: The Big Picture

- Problems with full joint distribution tables as our probabilistic models?
 - Too big to represent / estimate [Q: how many rows?]
- Bayes nets (or BNs aka graphical models):
 - Can compactly represent the joint distribution - as the product of the CPTs associated with each node (local interactions)
 - > Much more useful for practical time/space requirements & small data

Bayesian Networks

Definition (Bayesian Network)

A graph in which the following holds:

1. A set of random variables makes up the nodes in the network.
2. A set of directed links or arrows connects pairs of nodes.
3. Each node has a conditional probability table (CPT) that quantifies the effects the parents have on the node.
4. It is a directed, acyclic graph (DAG), i.e. no directed cycles.

BNs for applying Bayes' Theorem

An Olympic race winner tests positive for steroid use. Suppose:

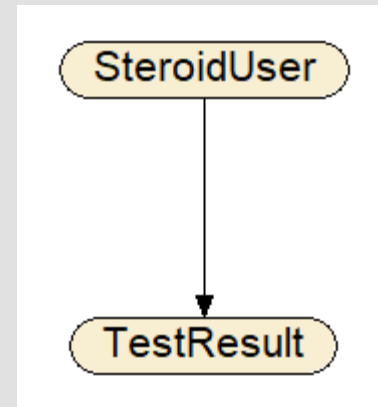
- 1% of all winning runners are known to take steroids.
- The probability of a positive result given that steroids are used is 95%.
- The probability of a false positive is 2%.

What is the (posterior) probability that the athlete did in fact take steroids, given the positive outcome of the test?

h = hypothesis (Athlete is taking steroids)

e = evidence (Positive test result)

$P(\text{SteroidUser}=\text{Yes})$	$P(\text{SteroidUser}=\text{No})$



Steroid User	$P(\text{Test}=\text{pos} \text{SU})$	$P(\text{Test}=\text{neg} \text{SU})$
Yes		
No		

BNs for applying Bayes' Theorem

An Olympic race winner tests positive for steroid use. Suppose:

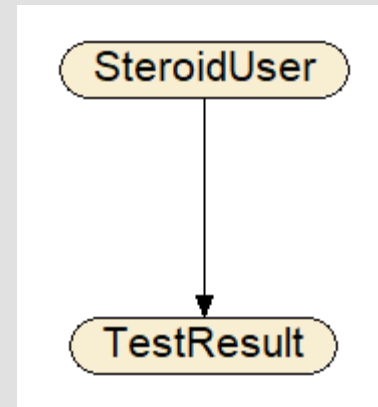
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What is the (posterior) probability that the athlete did in fact take steroids, given the positive outcome of the test?

h = hypothesis (Athlete is taking steroids)

e = evidence (Positive test result)

$P(\text{SteroidUser}=\text{Yes})$	$P(\text{SteroidUser}=\text{No})$
0.01	0.99



[View in Netica](#)

Steroid User	$P(\text{Test}=\text{pos} \text{SU})$	$P(\text{Test}=\text{neg} \text{SU})$
Yes	0.95	0.05
No	0.02	0.98

Example – Lung Cancer Diagnosis

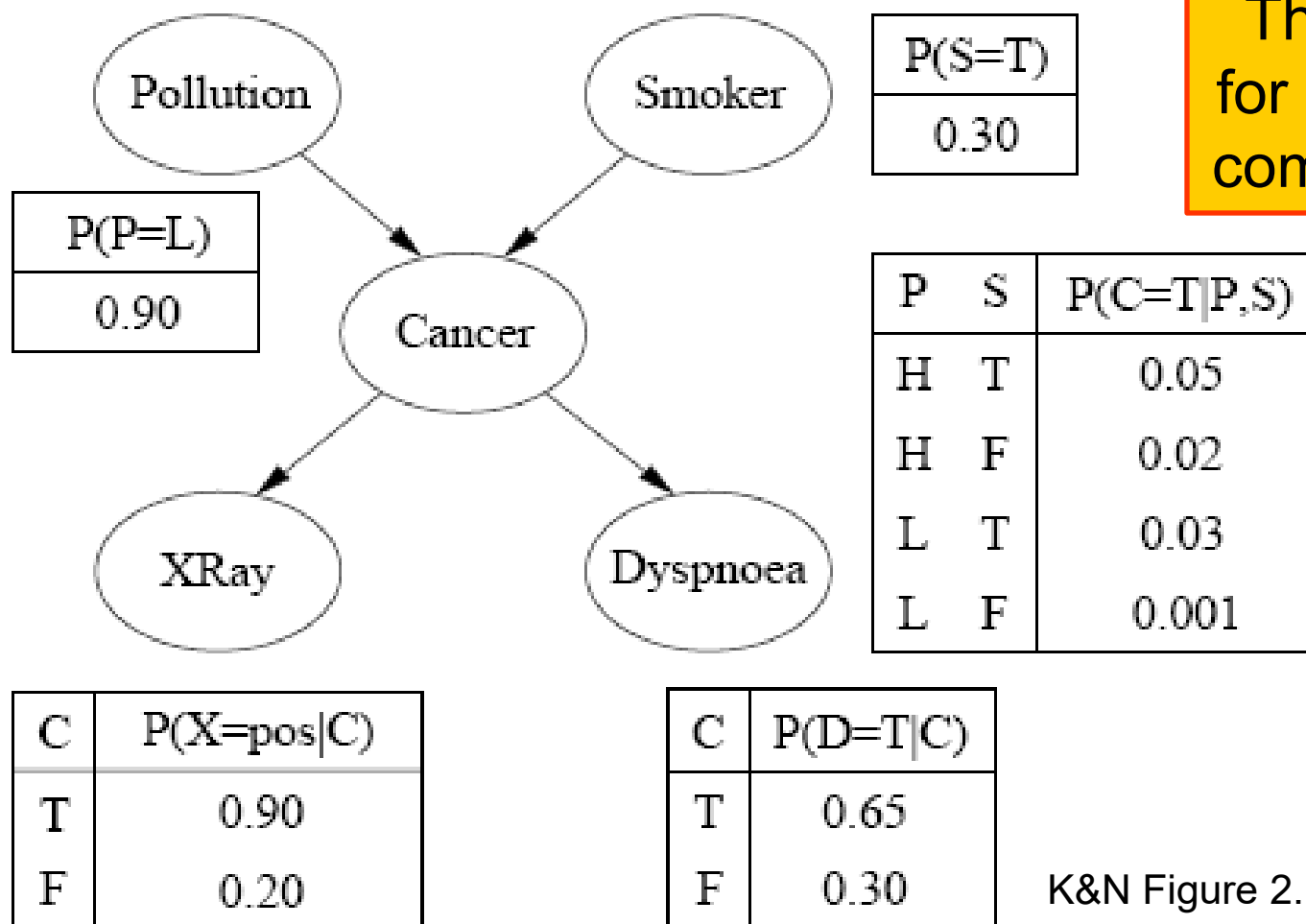
A patient has been suffering from shortness of breath (called dyspnoea) and visits the doctor, worried that they have lung cancer.

The doctor knows that other relevant information includes whether or not the patient is a smoker (increasing the chances of cancer) and what sort of air pollution they have been exposed to. A positive X-ray would indicate lung cancer.

(This example BN is taken from Korb&Nicholson, 2011 Section 2.2; it is a variant of the classic “Asia” BN, Lauritzen & Spiegelhalter, 1988, used in Neapolitan 1990, Netica tutorials, and so on)



Lung Cancer Example: CPTs



The value for =F is the complement

K&N Figure 2.1

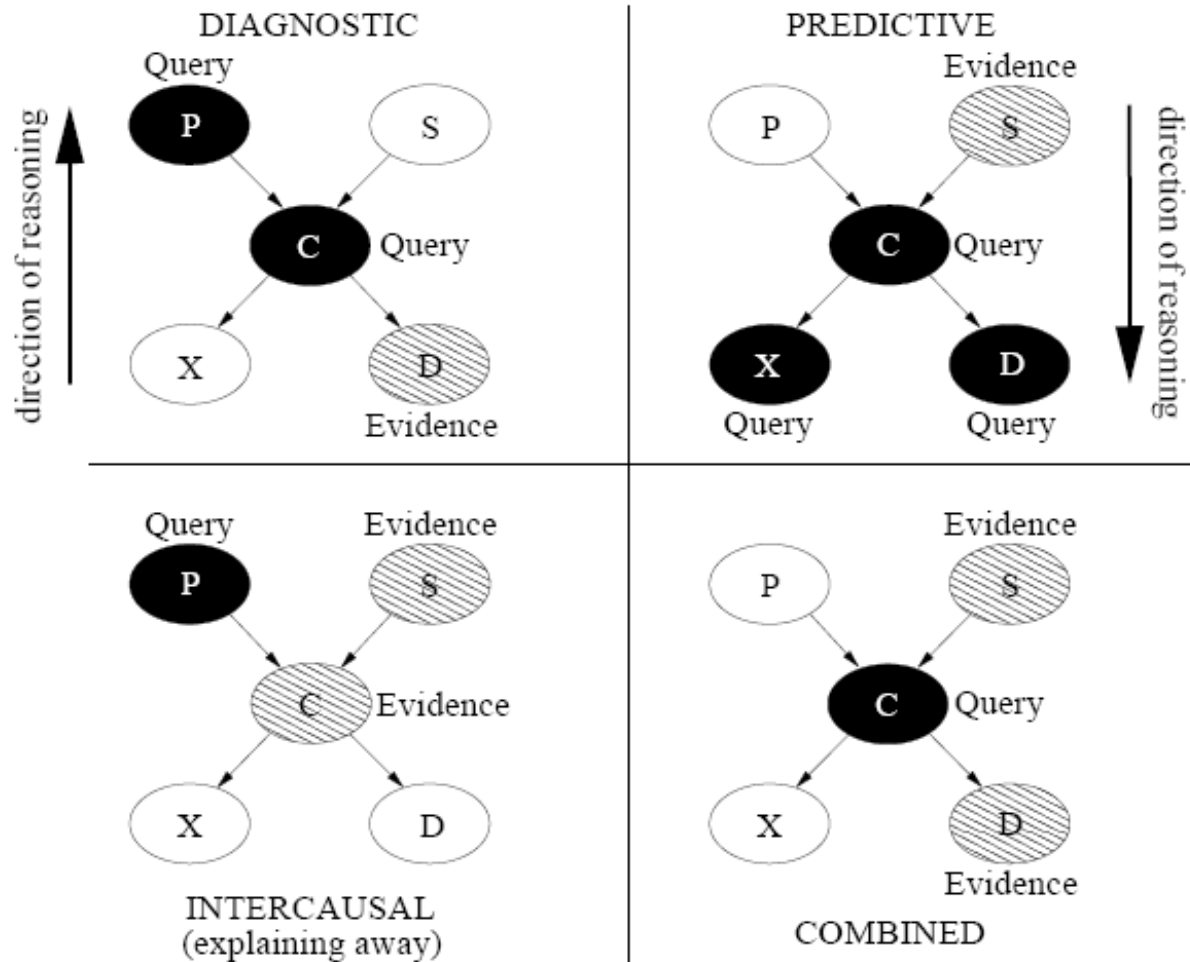
Reasoning with Bayesian Networks

- Basic task for any probabilistic inference system:
Compute the posterior probability distribution for a set of *query variables*, given new information about some *evidence variables*
- Also called *conditioning* or *belief updating* or *inference*

Lung Cancer Example –in Netica

- Nodes
- Arcs
 - What isn't represented in this model of the problem?
- CPTs
- Types of Reasoning:
 - Chain
 - Common Cause
 - Common Effect

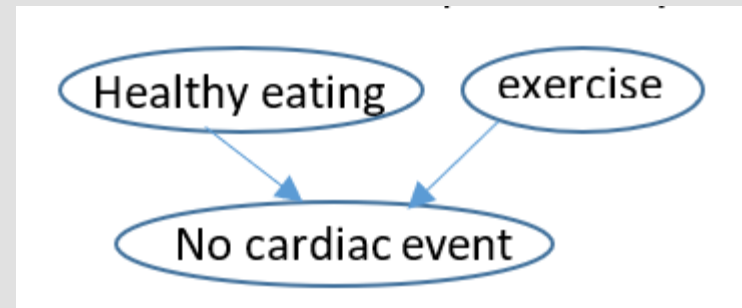
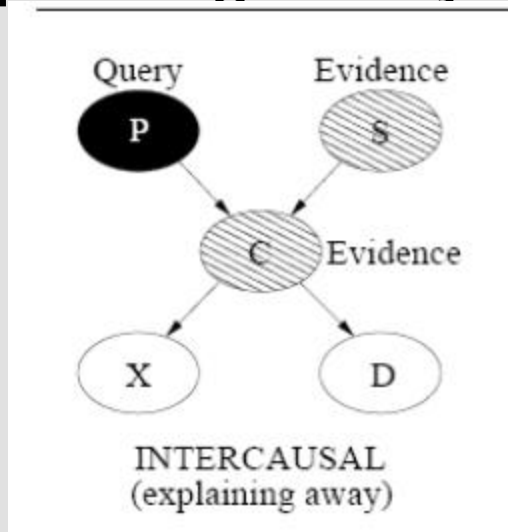
Types of Reasoning



K&N Figure 2.2

Notes on intercausal (explaining away)

- Many but not all common effect relationships are “explaining” away.



- (Noisy)-“Or” causes lead to explaining away.
- “And” causes do not (not an alternative explanation, but a supporting explanation)

Example: Native Fish

Example

A river with tree-lined banks contains native fish populations that need protection. The river can be affected by drought and passes through croplands, where pesticides are used. Rainfall helps native fish populations by maintaining water flow, but can also wash pesticides from the croplands into the river. There is concern that the trees and native fish will be affected by drought and pesticides.

(Nicholson et al., 2010)

(See http://bayesian-intelligence.com/publications/TR2010_3_NativeFish.pdf)

Bayesian Networks

Definition (Bayesian Network)

A graph in which the following holds:

1. A set of random variables makes up the nodes in the network.
2. A set of directed links or arrows connects pairs of nodes.
3. Each node has a conditional probability table (CPT) that quantifies the effects the parents have on the node.
4. It is a directed, acyclic graph (DAG), i.e. no directed cycles.

Nodes and Values

- A node (random variable) describes a part of the system that can be in several states, with each state having a certain probability.
- Nodes can be discrete **or** continuous.
 - Only discrete variables in this unit

Example: Native Fish

Example

Let's look at the nouns...

A river with tree-lined banks contains native fish populations that need protection. The river can be affected by drought and passes through croplands, where pesticides are used. Rainfall helps native fish populations by maintaining water flow, but can also wash pesticides from the croplands into the river. There is concern that the trees and native fish will be affected by drought and pesticides.

(Nicholson et al., 2010)

(See http://bayesian-intelligence.com/publications/TR2010_3_NativeFish.pdf)

Native Fish Scenario

Crops



Pesticide

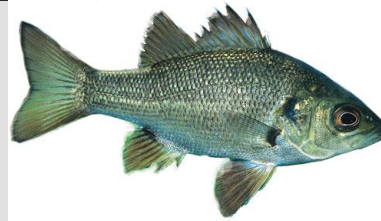


Trees



River

Water flow



Native fish



Native Fish example: nodes

Important query variables:

- Native fish abundance
- Tree condition

What have we left out (for now)?

- Crop, ???

Water related stressors:

- River flow
- Rainfall
- Drought conditions

Pesticide stressor:

- Pesticide use
- Pesticide concentration in river



Native Fish example: nodes

Node name	Type	Values
<i>Tree Condition</i>	Binary	{ <i>Good, Poor</i> }
	Ordered	{ <i>Good, Damaged, Dead</i> }
<i>Native Fish</i>	Ordered	{ <i>High, Medium, Low</i> }
<i>River Flow</i>		{ <i>High, Medium, Low</i> }
<i>Drought</i>		{ <i>Yes, No</i> }
<i>Pesticide Use</i>		{ <i>High, Low</i> }
<i>Pesticide in River</i>		{ <i>High, Low</i> }
<i>Rainfall</i>	1. Discretised ranges	{ <i>0-50, 51-200, 201-2000</i> }
	2. Ordered	{ <i>Below Average, Average, Above Average</i> }

Bayesian Networks

Definition (Bayesian Network)

A graph in which the following holds:

1. A set of random variables makes up the nodes in the network.
2. **A set of directed links or arrows connects pairs of nodes.**
3. Each node has a conditional probability table (CPT) that quantifies the effects the parents have on the node.
4. It is a directed, acyclic graph (DAG), i.e. no directed cycles.



Arcs = direct influence

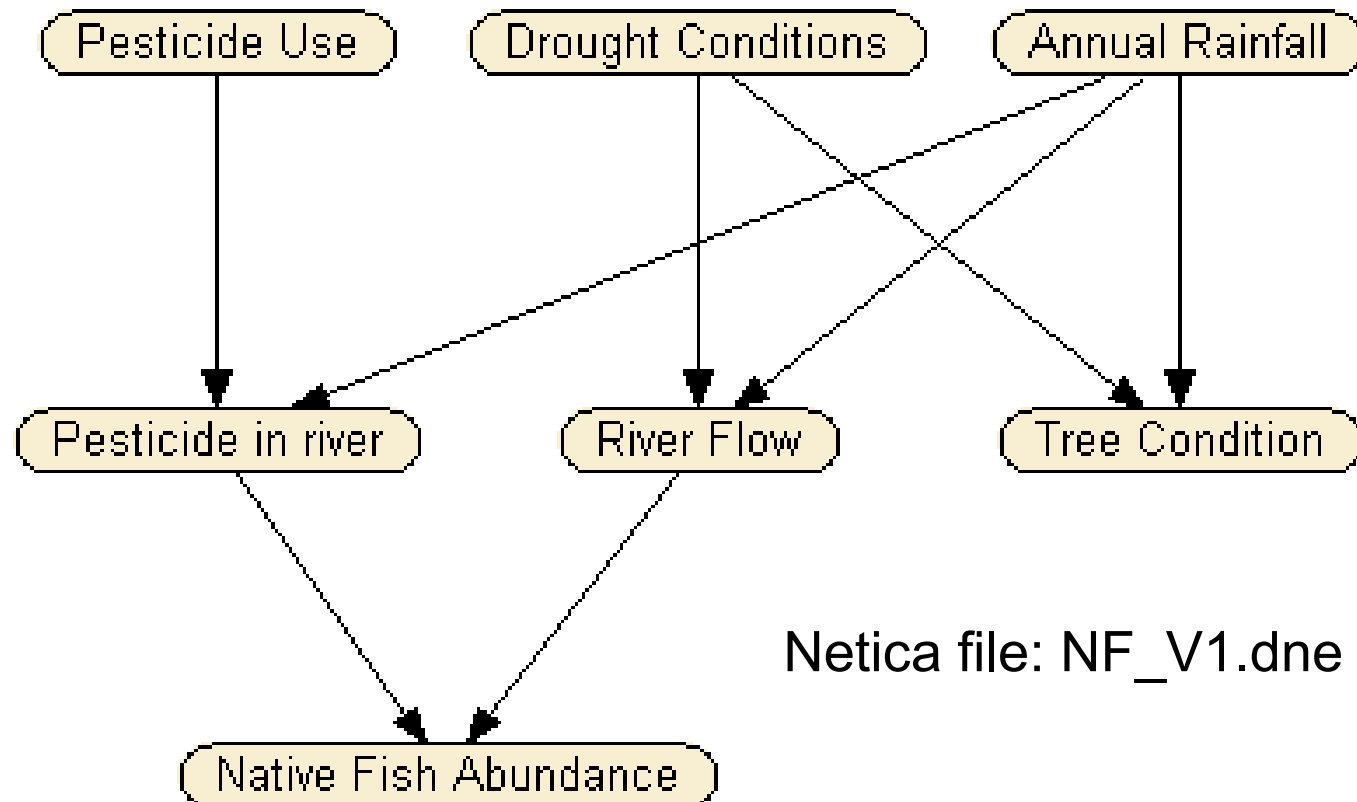
Terminology

$X \rightarrow Y$: "*X has a direct influence on Y*"

The nature of that influence may vary. It can just be statistical, but it often is (and ideally is for many reasons) causal.



Native Fish Example: Arcs



Native Fish Example: Arcs

What arcs have been left out?

i.e. What assumptions have been made?

- Pesticides don't affect tree condition
- No explicit representation of other causes of tree health or native fish abundance
 - > Summarised in uncertainty in links, e.g. from *Drought Conditions* to *Tree Conditions*
- Both laziness and ignorance in operation.

Structure Terminology

Terminology

*Family metaphor: Arcs go from **parent** nodes to **child** nodes.*

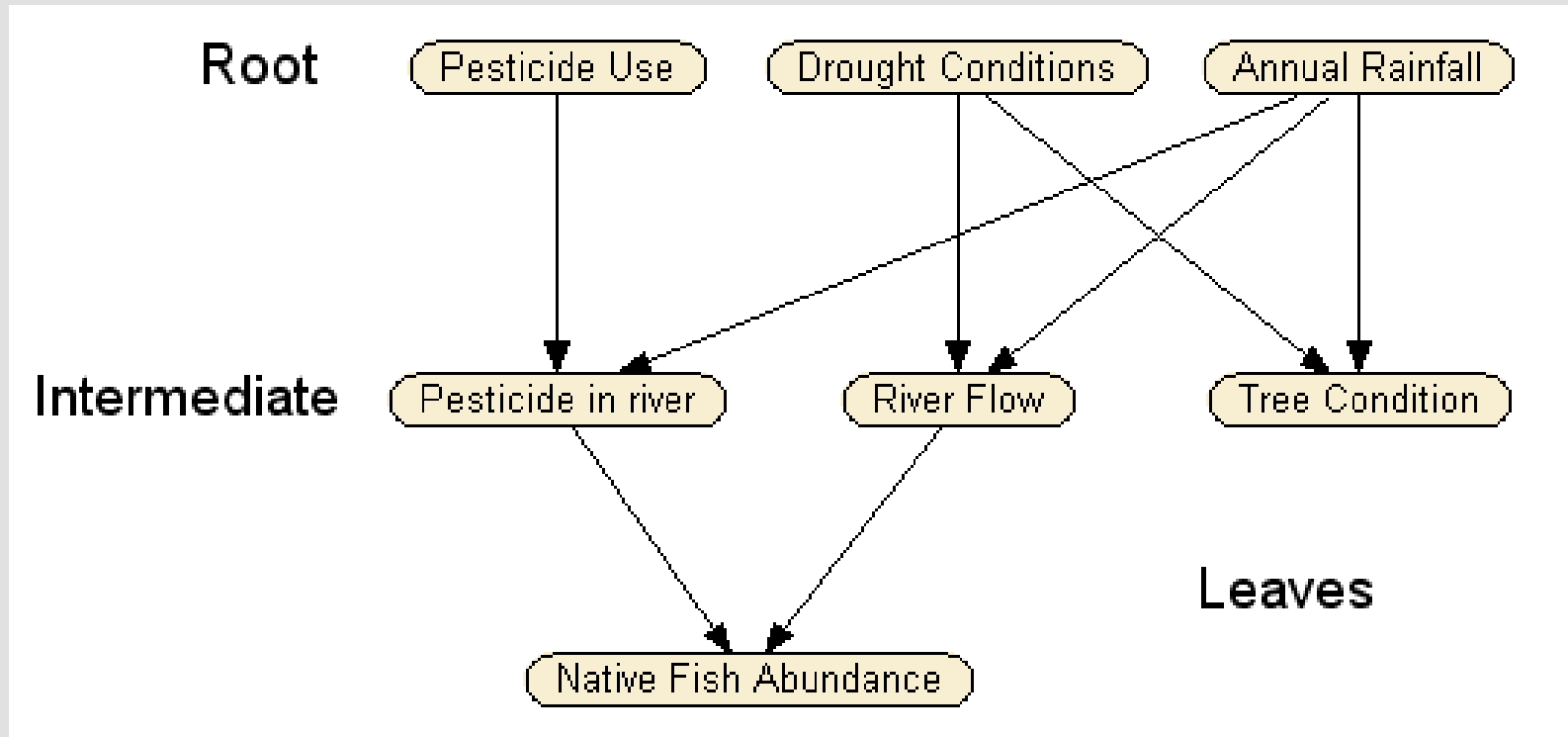
- Parent $\Rightarrow$ Child
- Ancestor $\Rightarrow \dots \Rightarrow$ Descendant

Tree analogy:

- **root** node: no parents
- **leaf** node: no children
- **intermediate** node: non-leaf, non-root

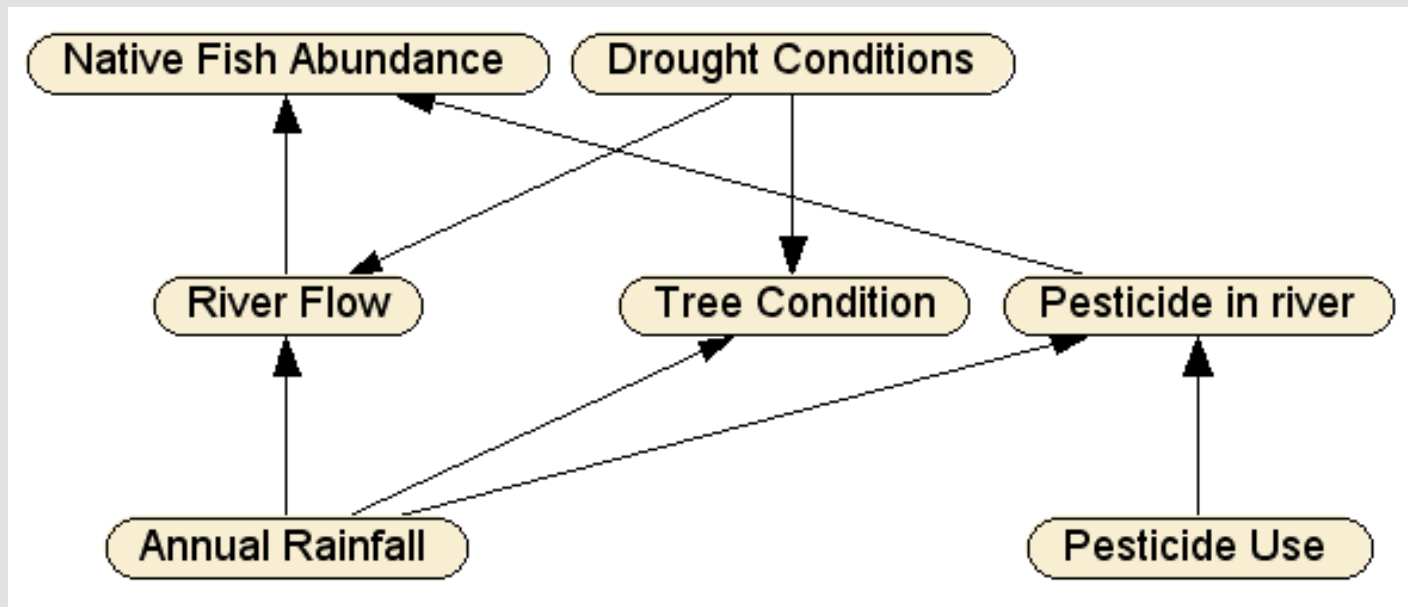


Structure terminology



Layout Convention

There are bad ways to lay out a BN...



Bayesian Networks

Definition (Bayesian Network)

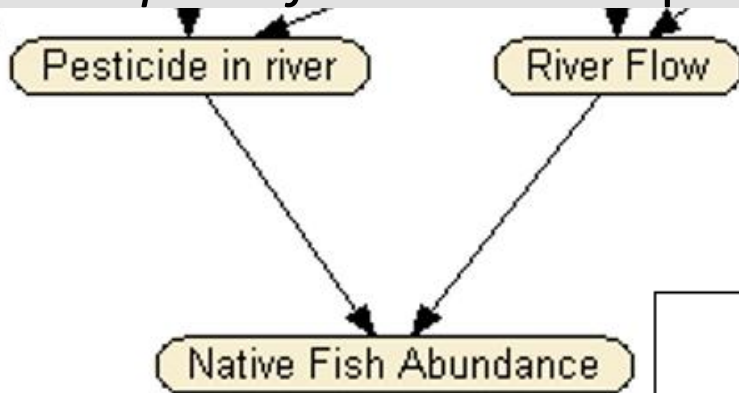
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Native Fish - CPTs

Once we've specified a topology, need to specify **conditional probability tables (CPT)** for each node

CPT *quantifies* the effect parents have on child node.



WORST CASE

BEST CASE

Pesticide in River	River Flow	P(FishAbundance PesticideInRiver, RiverFlow)		
		High	Medium	Low
High	Good	0.2	0.4	0.4
High	Poor	0.01	0.1	0.89
Low	Good	0.8	0.15	0.05
Low	Poor	0.05	0.15	0.8

Each row sums to 1

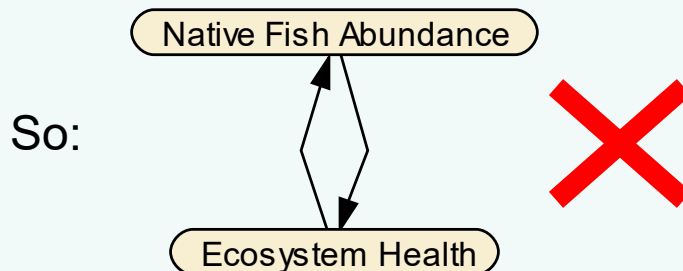


Bayesian Networks

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4. **It is a directed, acyclic graph (DAG), i.e. no directed cycles.**



But in practice, this is a very common extension (Dynamic BNs)

Representing the joint probability distribution

To calculate the joint from a Bayesian network, can use chain rule plus the Markov property

First, order nodes roots to leaves (topological order): $x_1, \dots, x_n$

Chain rule:

$$\Pr(x_1, \dots, x_n) = \prod_{i=1}^n \Pr(x_i | x_1, \dots, x_{i-1})$$

Markov property of BNs/DAGs:

$$\Pr(x_i | \text{parents}(x_i), \{x_j\}_{j < i}) = \Pr(x_i | \text{parents}(x_i))$$

(i.e., every node is independent of all non-descendants given its parents)

Combining, gives:

$$\Pr(x_1, x_2, \dots, x_n) = \prod_{i=1}^n \Pr(x_i | \text{parents}(X_i))$$



Representing the joint probability distribution

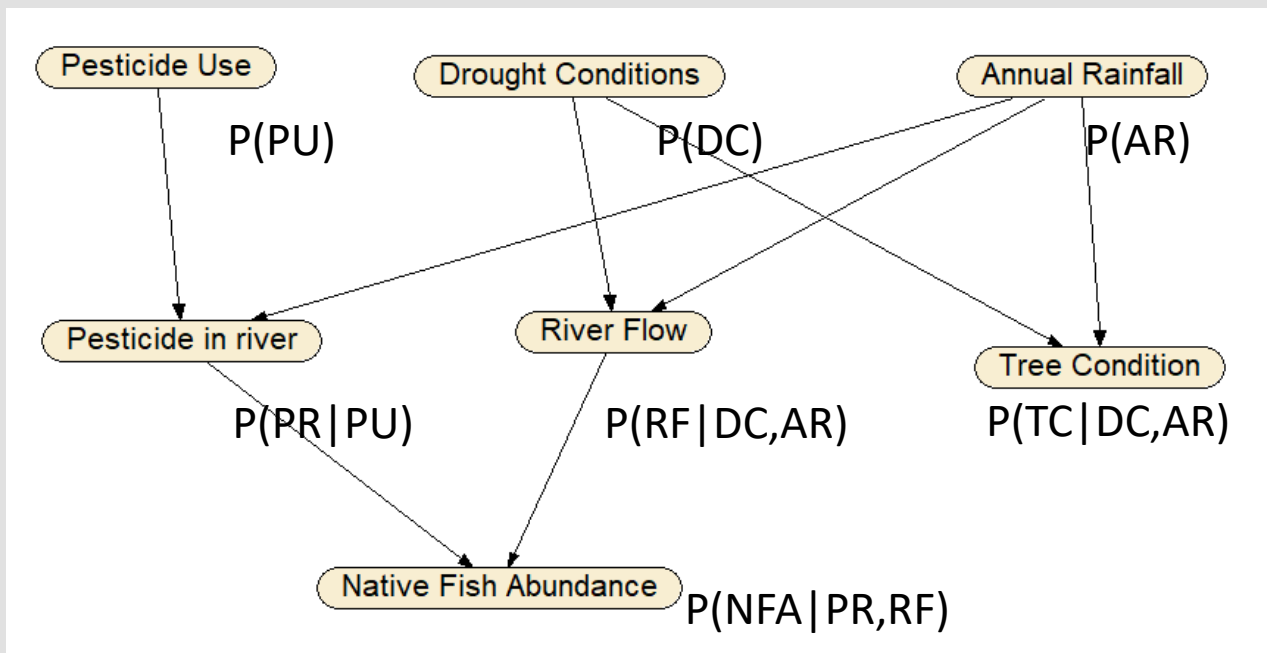
$$\Pr(x_1, x_2, \dots, x_n) = \prod_{i=1}^n \Pr(x_i \mid \text{parents}(X_i))$$

Native Fish joint distribution: (432 rows in total)

PU	AR	DR	PR	RF	TC	FA	P()
High	AboveAverage	Yes	High	Good	Good	High	0.173%
High	AboveAverage	Yes	High	Good	Good	Medium	0.346%
High	AboveAverage	Yes	High	Good	Good	Low	0.346%
High	AboveAverage	Yes	High	Good	Dam.	High	0.346%
High	AboveAverage	Yes	High	Good	Dam.	Medium	0.691%
High	AboveAverage	Yes	High	Good	Dam.	Low	0.691%
...							
Low	BelowAverage	No	Low	Poor	Dead	High	0.001%
Low	BelowAverage	No	Low	Poor	Dead	Medium	0.003%
Low	BelowAverage	No	Low	Poor	Dead	Low	0.016%

Representing the joint probability distribution

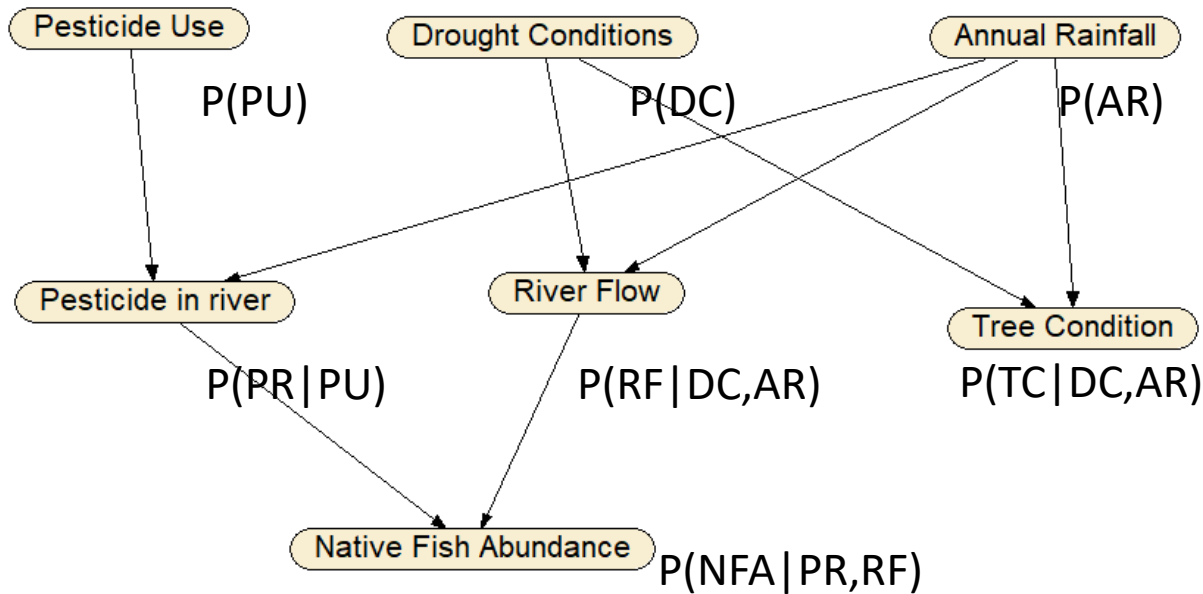
$$\Pr(x_1, x_2, \dots, x_n) = \prod_{i=1}^n \Pr(x_i \mid \text{parents}(X_i))$$



For Native Fish = ???

Representing the joint probability distribution

$$\Pr(x_1, x_2, \dots, x_n) = \prod_{i=1}^n \Pr(x_i \mid \text{parents}(X_i))$$



BN parameters:
2+2+3

+12+12+18

+12 = 61 (36 free)

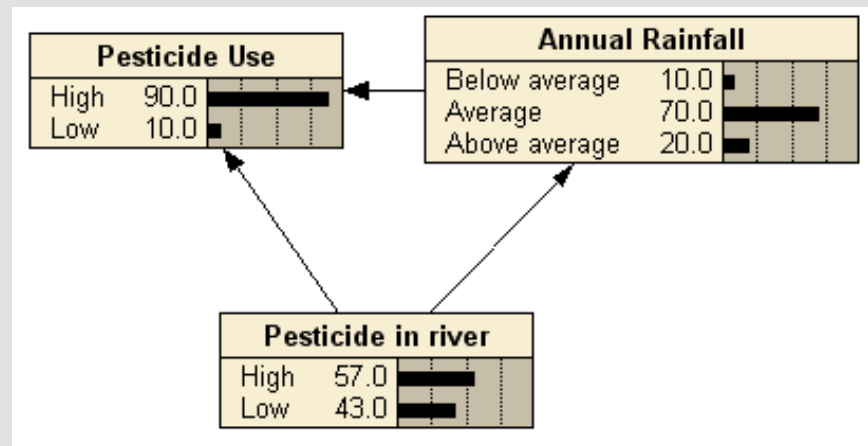
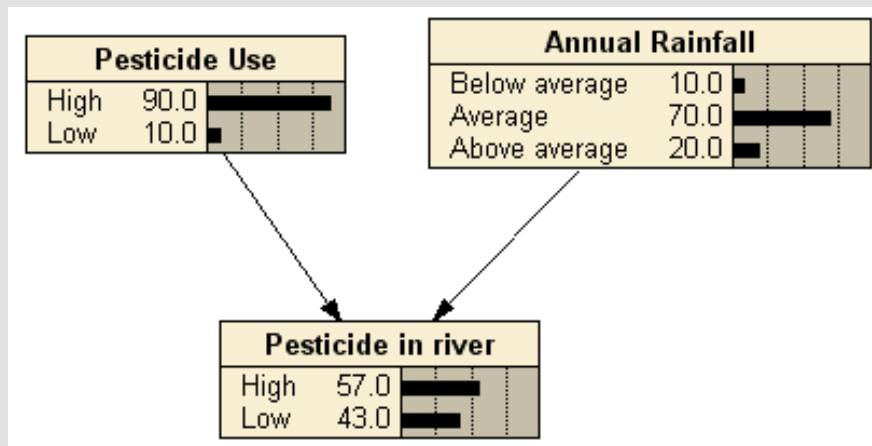
Joint has: 432 (431 free)

→ $P(\text{PU}) \times P(\text{DC}) \times P(\text{AR}) \times P(\text{PR} \mid \text{PU}) \times$
 $P(\text{RF} \mid \text{DC}, \text{AR}) \times P(\text{TC} \mid \text{DC}, \text{AR}) \times P(\text{NFA} \mid \text{PR}, \text{RF})$



Representing the joint distribution

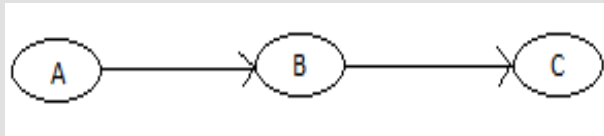
These BNs both represent the same joint distribution



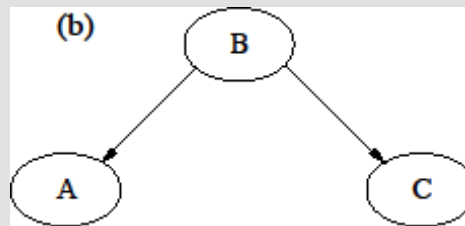
(Note: implications for learning BN structure)

Representing dependence/independence relationships

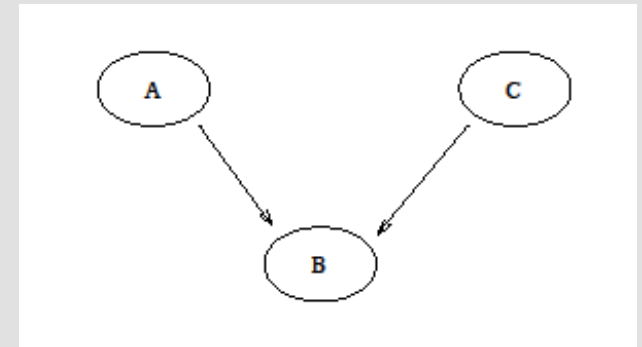
Bayesian networks are constructed from 3 basic fragments:



(a)
Chain



(b)
Common
Cause



(c)
Common
Effect

Conditional Independence: Causal Chains

Causal chains give rise to conditional independence:

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$$P(C|A \wedge B) = P(C|B)$$

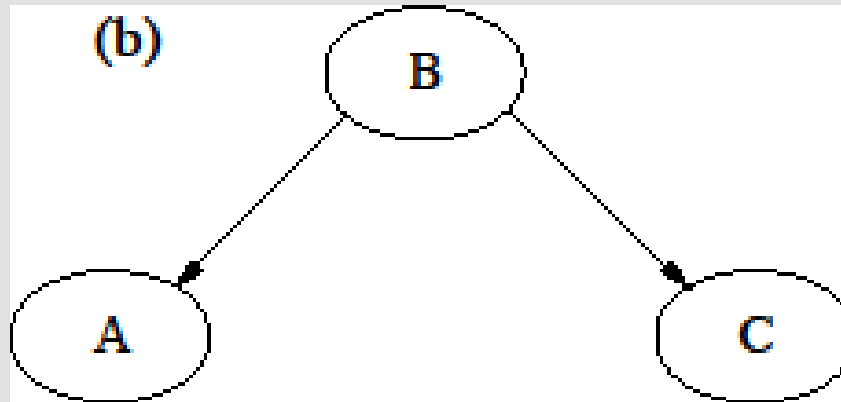
Example: Pesticide use leads to pesticide concentration in the river which impacts on native fish abundance

$$P(C|A \wedge B) = P(C|B)$$

Example: Pesticide use leads to pesticide concentration in the river which impacts on native fish abundance

Conditional Independence: Common Causes

Common causes (or ancestors) also give rise to conditional independence:

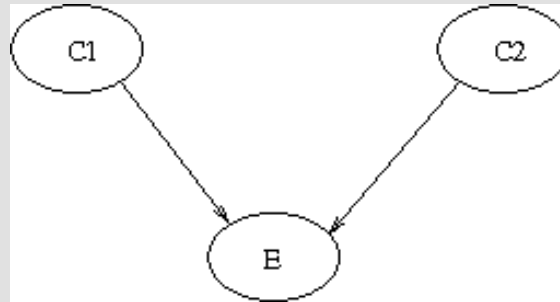


$$P(C|A \wedge B) = P(C|B)$$

Example: General drought condition directly influences both tree condition and river flow

Conditional Dependence: Common Effects

Common effects (or their descendants) give rise to conditional *dependence*:



$$P(C1|C2 \wedge E) \neq P(C1|E) \equiv \neg(C1 \perp\!\!\!\perp C2 | E)$$

$$\text{Yet: } P(C1|C2) = P(C1) \equiv C1 \perp\!\!\!\perp C2$$

Native fish abundance is a common effect of pesticide concentrations in the river as well as river flow.

Given low native fish abundance, high pesticide concentration “explains away” low river flow.

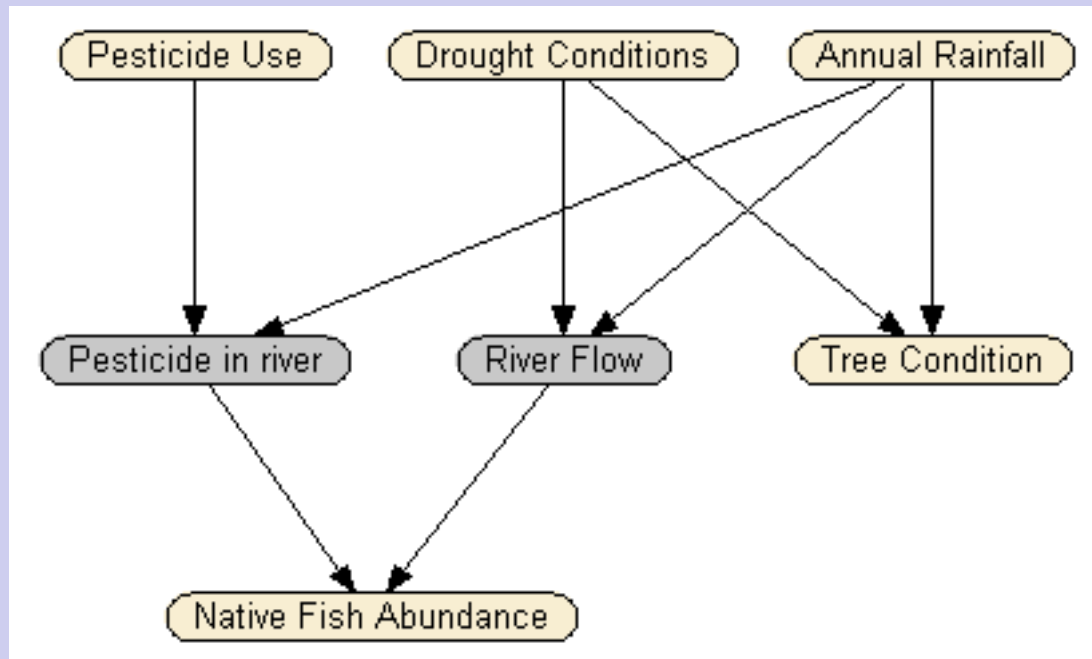
Dependence and Independence?

What do these mean in practice when modelling with BNs?

- Given evidence about some nodes, we can determine whether other nodes are independent of each other (“separated”)

Dependence and Independence?

Example (Independence)



Given evidence about Pesticide in river and River Flow, Native Fish Abundance is now “separated” from the other nodes.

Summary: Bayesian Networks

- Bayes nets compactly encode joint distributions
- BNs are a natural way to represent conditional independence information
 - qualitative: links between nodes – independencies of distributions can be deduced from a BN graph structure (can determine: see d-separation next week)
 - quantitative: conditional probability tables (CPTs)
- BN reasoning
 - computes the probability of query variables given evidence variables
 - is flexible – we can enter evidence about any node and update beliefs in other nodes

Reading and Software

- Reading

- Russell, S. and Norvig, P. (2010), *Artificial Intelligence – A Modern Approach* (3rd ed), Prentice Hall
 - > Chapters 14.1-14.4.1
- Korb, K. and Nicholson, A. (2010), *Bayesian Artificial Intelligence* (2nd ed), Chapman and Hall
 - > Chapters 1 & 2

- Software

- Netica – <http://www.norsys.com/>



Next Lecture

- Week 6 – Bayesian Networks II
 - d-separation (graphical independence)
 - BN inference algorithms (overview)
 - Bayesian **decision** networks
 - Knowledge engineering Bayesian networks
 - Case studies: COVID-19 causal BNs for
 - > COVID testing
 - > Patient prognosis / clinical decision support